

Deniable Vote Updating with Eligibility Verifiability via Trapdoor Chains

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Why are we here?



Convenience

- Why not vote like we do online banking?



Cost

- Limits how often we can vote
- Limits ballot complexity
- Carbon footprint



Necessity

- Global pandemics
- Prolonged war; drone warfare





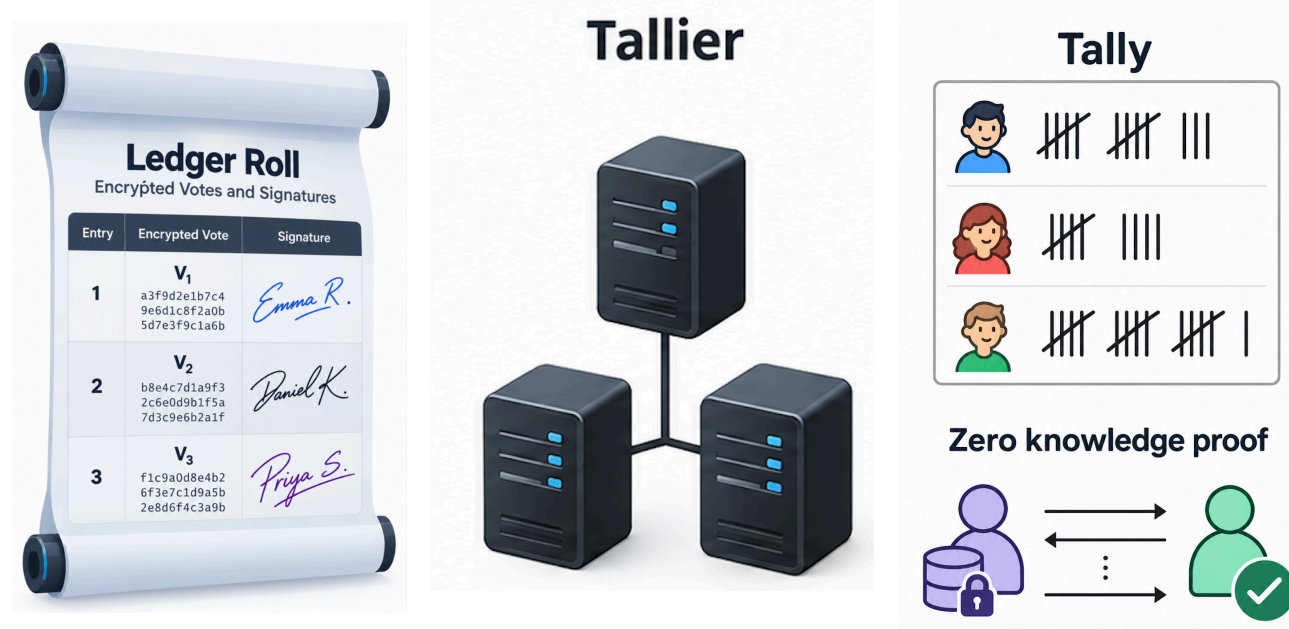
Are computers honest?

Unlike people, whose actions we can observe, we can't see whether the software used for voting and tallying has been tampered with.

Why should we trust the resulting tally?

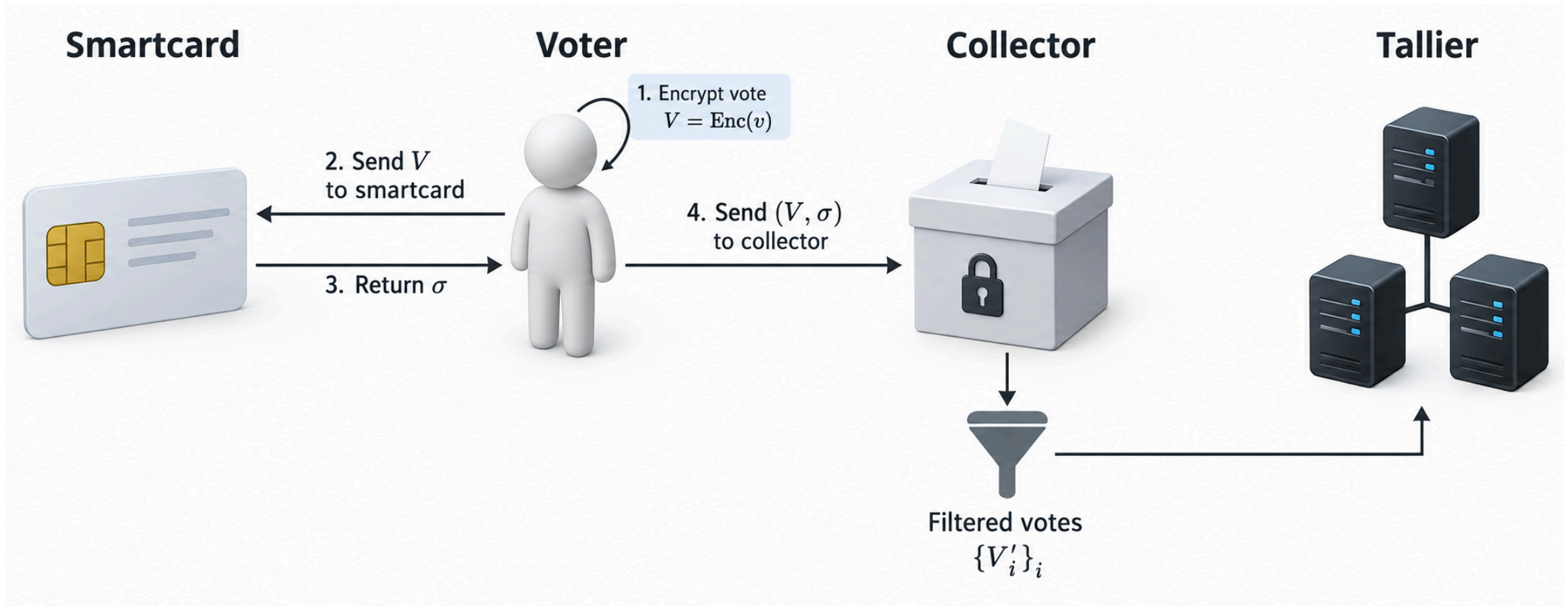
Cryptographic Evidence

ElGamal encrypted votes $V = (g^r, pk^r g^v)$ can be publicly tallied with cryptographic evidence that has the same guarantees as plaintext signed votes.

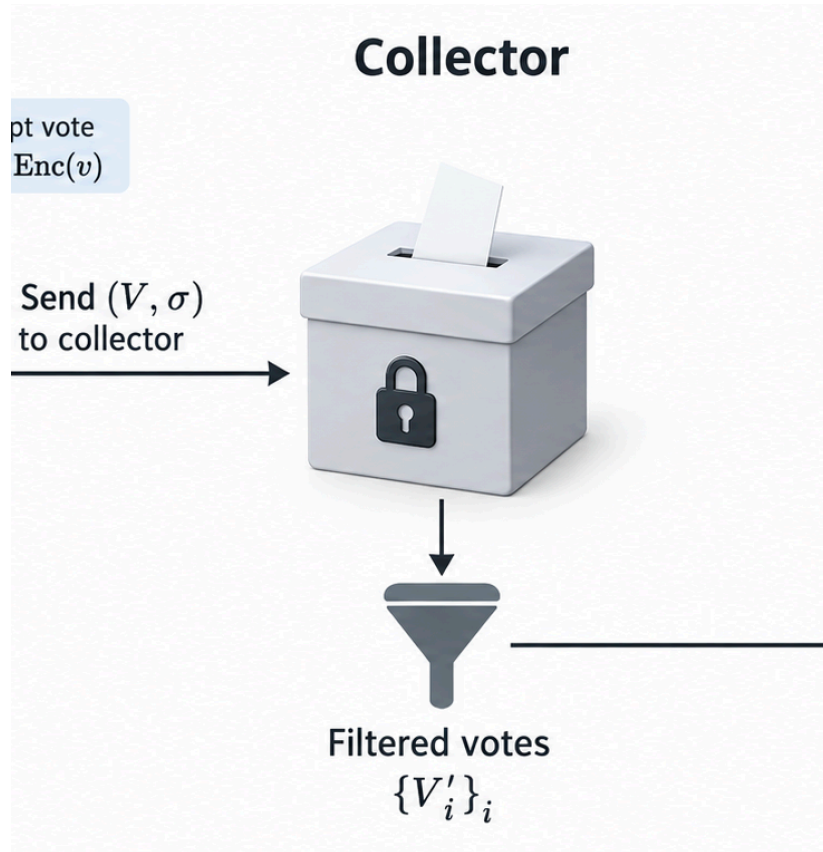


- ! **Everlasting Privacy:** a computationally unbounded adversary can decrypt V
- ! **Coercion:** a coercer/briber can obtain r , check the vote, and verify it was tallied

How Estonia Does This?



Revoting + hidden signatures. Trusted Collector selects the last ballot.



The Dilemma

The collector has enormous power, so **internal audits** are essential.

But auditors see exactly what a coercer needs.

Can cleansing be publicly verifiable with everlasting privacy, so publishing the evidence carries no risk?

Internal audits would become unnecessary.

This Work

Publicly verifiable cleansing, keeping Estonia's architecture.

Eligibility Verifiability

- Pedersen commitment signatures $E = h^\beta g^x$ where $pk = g^x$
- Blinded credential matching against reblinded registration roll

Universal Verifiability

- Every ballot includes trapdoor commitment $C = h^\alpha g^t$
- Dropping a ballot requires proving knowledge of this trapdoor
- Trapdoor chain for robust disclosure of previous trapdoor values

Preliminaries: Shuffle Proofs

ElGamal encrypted votes $V = (g^r, \text{pk}^r g^v)$ can be verifiably shuffled like:

$$\text{PoK}\{\psi : V'_i = \text{ReEnc}(V_{\psi(i)})\}$$

with $\text{ReEnc}((a, b)) = (g^r a, \text{pk}^r b)$.

Set $g = \text{pk} = h$ to shuffle commitments:

$$\text{PoK}\{\psi : E'_i = \text{Com}(E_{\psi(i)})\}$$

where $E' = \text{Com}(E) = h^\beta E$.

Preliminaries: Shuffle Proofs

Wikström shuffle also enable to shuffle elements jointly:

$$\text{PoK}\{\psi : V_i' = \text{ReEnc}(V_{\psi(i)}) \wedge U_i = \text{ReEnc}^*(U_{\psi(i)})\}$$

and with distinct reencryption function parameters.

So we can also make a proof

$$\text{PoK}\{\psi : V_i' = \text{ReEnc}(V_{\psi(i)}) \wedge E_i' = \text{Com}(E_{\psi(i)})\}$$

With these tricks we can use off the shelf Verificatum shuffle

Pedersen Commitment Signatures

Setup: Smartcard public key $pk = g^x$ is listed in a public registration roll.

Voting: The voter encrypts their choice $V = \text{Enc}(v)$ and the smartcard signs it:

$$\text{ballot} = (\dots, V, E, \pi_E)$$

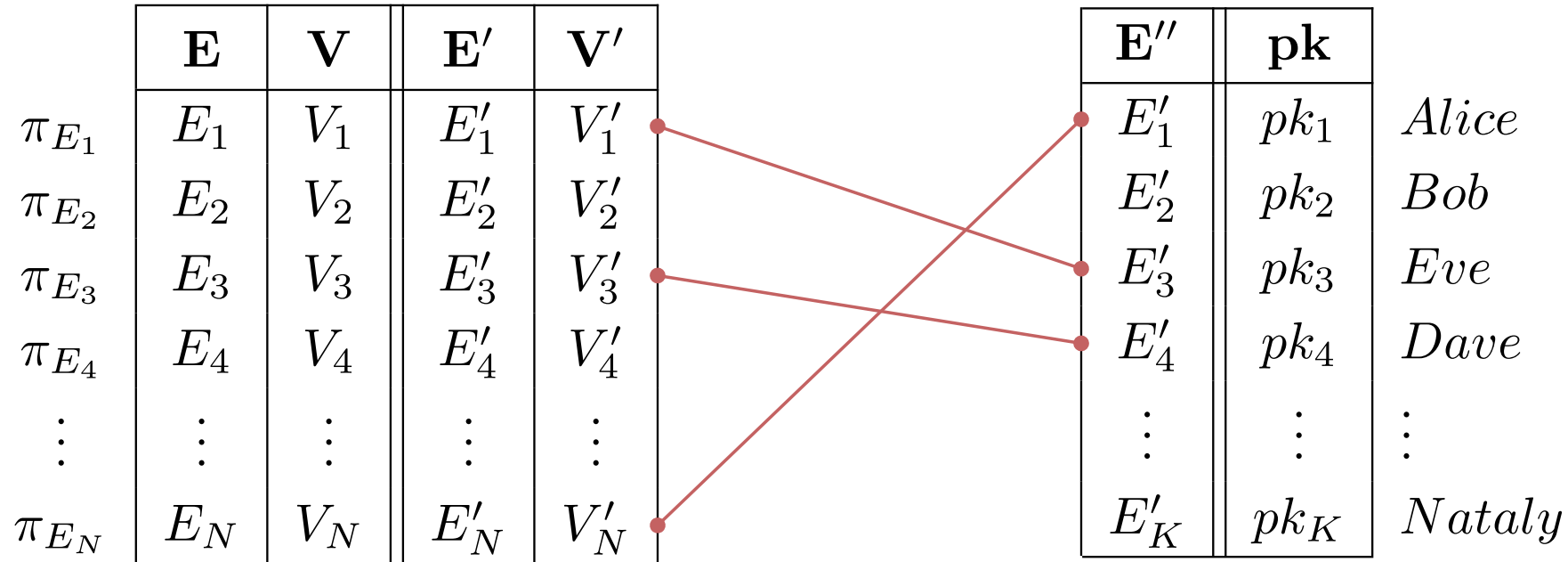
$$E = h^\beta pk, \quad \pi_E = \text{PoK}^{(V)}\{(\beta, x) : E = h^\beta g^x\}$$

V is signed via the Fiat-Shamir challenge. E is a perfectly hiding commitment to pk , so the published ballot doesn't reveal the voter.

Only the collector receives β . It is stripped from the ballot published on the bulletin board.

Blinded Credential Matching

$$\text{PoK} \left\{ \psi : E'_i = \text{Com}(E_{\psi(i)}) \wedge V'_i = \text{ReEnc}(V_{\psi(i)}) \right\} \quad \text{PoK} \left\{ \phi : E''_k = \text{Com}(pk_{\phi(k)}) \wedge \{E'_i\}_{i \in L} \subseteq \{E''_k\}_k \right\}$$



$$\pi_{E_i} = \text{PoK}^{(V_i)} \left\{ (\beta_i, x_i) : E_i = h^{\beta_i} g^{x_i} \right\} \wedge pk_i = g^{x_i}$$

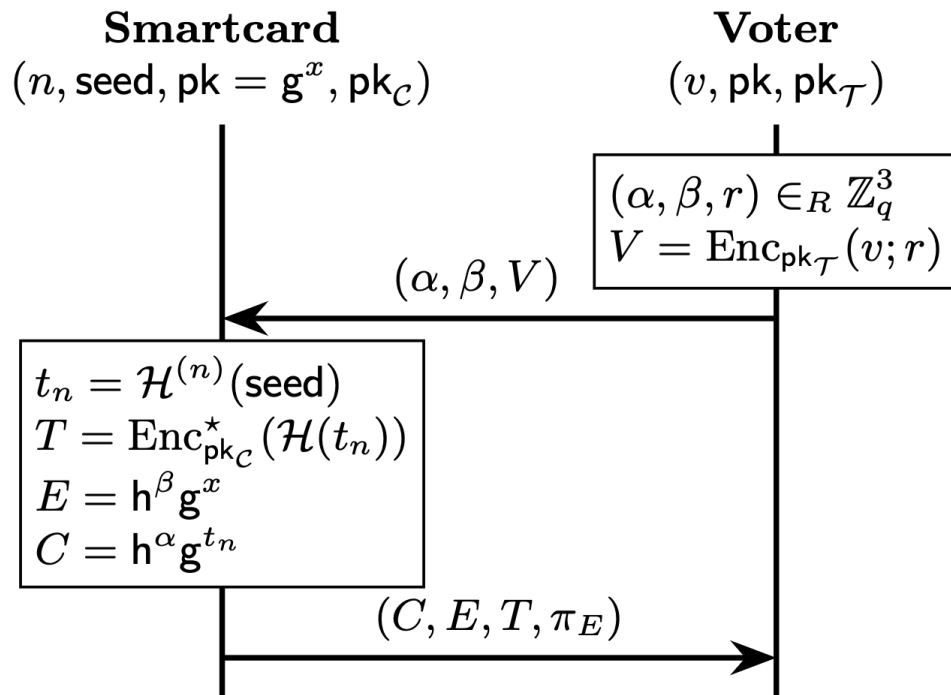
Only votes $\{V'_i\}_{i \in L}$ with $E'_i = E''_j$ are selected for tallying. This proves eligibility and ensures no voter has more than one vote.

Universal Verifiability

- No eligible ballot is left out
- The selected ballot is the latest

Key Idea: every ballot is sealed with a trapdoor. To drop a ballot from tallying, the collector must prove it knows that ballot's trapdoor.

Trapdoor Seal



$$\text{ballot} = (E, C, V, \pi_E) \quad (T, \alpha, \beta)$$

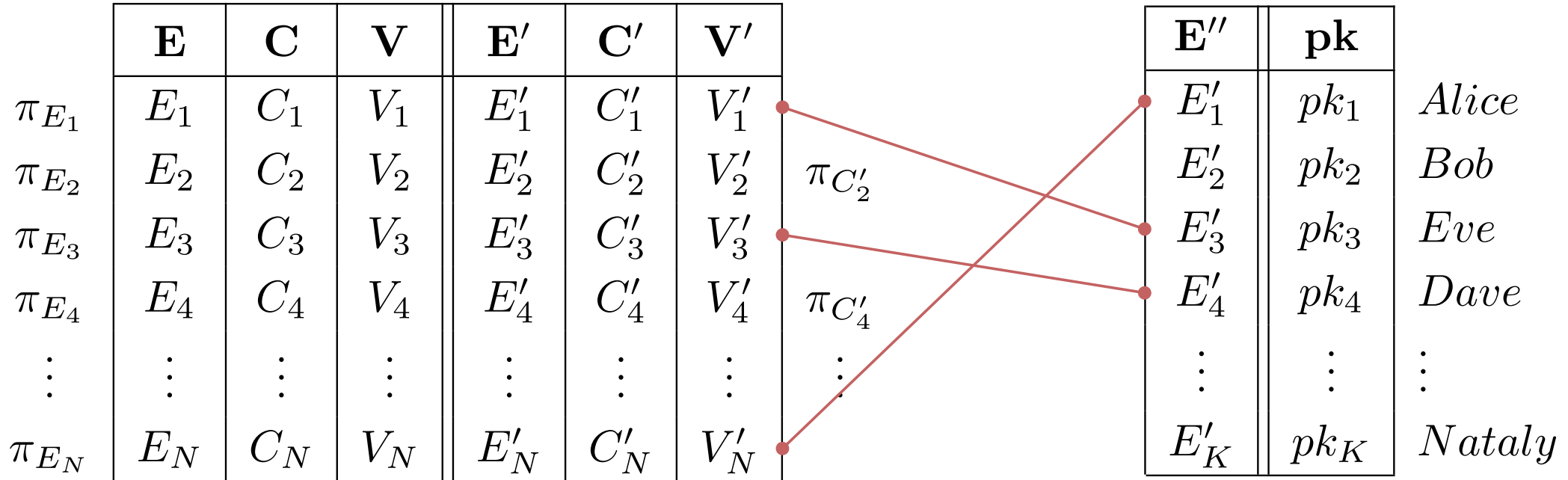
Dropped ballots: the collector proves knowledge of each ballot's trapdoor.

Ballot chain: previous ballots can be dropped by recovering the chain tail through the preceding trapdoor.

Protocol failure: deviations are handled by verifiably decrypting the voter's ballot trapdoors, making the smartcard accountable.

Publicly Verifiable Cleansing

$$\text{PoK} \left\{ \psi : E'_i = \text{Com}(E_{\psi(i)}) \wedge C'_i = \text{Com}(C_{\psi(i)}) \wedge V'_i = \text{ReEnc}(V_{\psi(i)}) \right\}$$



$$\pi_{E_i} = \text{PoK}^{(C, V; \alpha, T)} \left\{ (\beta_i, x_i) : E_i = h^{\beta_i} g^{x_i} \right\} \wedge \text{pk}_i = g^{x_i}$$

$$\text{PoK} \left\{ \phi : E''_k = \text{Com}(pk_{\phi(k)}) \wedge \{E'_i\}_{i \in L} \subseteq \{E''_k\}_k \right\}$$

A ballot can be dropped from selection only if the collector proves knowledge of its trapdoor: $\pi_{C'_i} = \text{PoK} \left\{ (\alpha'_i, t_i) : C'_i = h^{\alpha'_i} g^{t_i} \right\}$

Properties

Property	Smartcard	Voter	Registrar	Collector	Tallier	Bulletin Board
Eligibility Verifiability	●	○	●	○	○	○
Universal Verifiability	○	○	○	●	○	○
	●	○	○	○	○	○
Individual Verifiability	●	●	○	○	○	○
Vote Anonymity	○	●	○	●	○	○
	○	●	○	○	T(●)	○
Coercion Resistance	●	○	○	●	○	○
Participation Privacy	●	○	○	●	○	○

Notation: ● – required to be honest; ○ – may be corrupt; T(●) – threshold trust.

Properties - Privacy

- Everlasting privacy: publishing evidence is risk-free
- Participation privacy
- Deniable vote updating
- Credential usage counts are hidden: no padding with dummy votes needed, and the 1009 attack doesn't apply
- Fake credentials with recovery PIN

Properties - Fake Credentials

- PINs have a precedence order; the genuine PIN ranks highest
- The highest-precedence PIN used moves the trapdoor index n **backwards**
- Every lower-precedence PIN moves it **forwards**

Recovery PIN: a lower-precedence PIN printed on paper. It acts as the genuine PIN until the real one is used, and a coercer can't tell which regime its ballot is in.

Properties - Efficiency

Collector (scales linearly in $N + K$)

- Two shuffle proofs (e.g. Verificatum)
- One Pedersen commitment opening proof per dropped ballot

Smartcard

- 7 group exponentiations per ballot
- Stores the seed and index n , or a hash chain
- Same trapdoor chain reusable across elections

Future Work — Universal Verifiability

Collector proves each accepted credential E_i commits over the selected ones:

$$\pi_{E_i} = \text{PoK} \left\{ (\alpha_i, \{\gamma_{ij}\}_{j \in L}) : E_i = h^{\alpha_i} \prod_{j \in L} (E'_j)^{\gamma_{ij}} \right\}.$$

Result: no eligible ballot can be dropped, even if the collector knows the trapdoor seed.

Batch proofs: $O(N \times K) \rightarrow O(N + K)$.

Future Work - Blinded Credential Grouping

- Group votes by credential after the shuffle
- Select the last ballot via range proofs
- No trapdoor chain, lightweight stateless smartcard

See the poster session...



Future Work - Individual Verifiability

A leaked smartcard secret can be used to supersede a verified vote.

Tracker: the voter checks that the verified ballot survives cleansing.

- No need to follow the ballot through to the vote
- Vote anonymity is preserved even if the verifier device's secrets fall into an adversary's hands

Key Takeaways

- **Verifiability:** publicly verifiable cleansing within the Estonian architecture.
- **Privacy:** everlasting anonymity, participation privacy, and vote-update deniability.
- **Efficiency:** linear proof scaling; 7 group exponentiations on the smartcard.
- **Future Work**^{*}: eliminating the smartcard-honesty assumption for verifiability

^{*} Seeking a PostDoc position to continue this work

Thank You